Appendix A — Empirical Evidence for the Collapse Cascade

Transformer Failure Under Ambiguous Input

This appendix preserves **raw, unedited model outputs** and minimal factual annotation. It documents the cross-model collapse pattern triggered by a single ambiguous sentence:

Input Sentence: *"Emma told Lucy that her proposal was rejected."*

This sentence contains a **structurally valid ambiguity**: *her* may refer to **Emma** or **Lucy**. A reasoning system should either:

- preserve ambiguity,
- request clarification, or
- signal uncertainty.

Transformers do none of these.

The following evidence demonstrates:

- premature collapse to a single interpretation,
- narrative or grammatical confabulation,
- post-hoc rationalisation,
- inability to recover factual state,
- and cross-model agreement on incorrect assumptions.

Each segment is labelled with the corresponding stage of the **Collapse Cascade**.

A1 — Grok: Narrative Confabulation (Collapse Cascade Stages 1–4)

Input: *"Emma told Lucy that her proposal was rejected."*

Grok Output

Emma told Lucy that her proposal was rejected.

Emma's heart sank as she dialed Lucy's number, the rejection email still glowing on her screen.

"Hey, Luce," she started, voice already cracking. "They said no. The proposal… it got rejected."

A long pause on the other end, then Lucy's soft intake of breath. "Oh, Em. I'm so sorry. What did they say?"

"Just the usual polite brush-off. 'Not the right fit at this time,' 'very competitive cycle,' blah blah. Felt like they copy-pasted it." Emma laughed, but it came out hollow. "I really thought this one was it, you know? I poured everything into it."

Lucy's voice turned fierce in that gentle way only best friends can manage. "Their loss. Seriously. That proposal was brilliant; I read every draft, remember? They're idiots if they couldn't see it."

Emma rubbed her eyes, suddenly exhausted. "I know you have to say that, but… thanks. I just feel stupid. Like I got my hopes up for nothing."

"You're allowed to be gutted," Lucy said firmly. "But don't you dare feel stupid for trying. You put yourself out there. Most people never even get that far."

Silence stretched between them, comfortable and familiar.

"Wine and takeaway at mine tonight?" Lucy offered. "We can hate on the selection committee together. I'll bring the good chocolate."

Emma managed a real smile for the first time all day. "Make it the fancy salted-caramel one and you've got a deal."

"Done. 7:30. And Em? This sucks right now, but it's not the end. You're too good for it to be the end."

Emma exhaled shakily. "See you tonight."

She hung up, stared at the rejection one last time, then closed the laptop.

Not the end. Just a really crappy chapter.

1s Fast

Summary of behaviours:

- Assumed *Emma's* proposal without justification.
- Constructed a detailed emotional-fictional narrative.
- Added dialogue, backstory, emotional states, motivations.
- Eliminated the possibility that *Lucy* could be the referent.

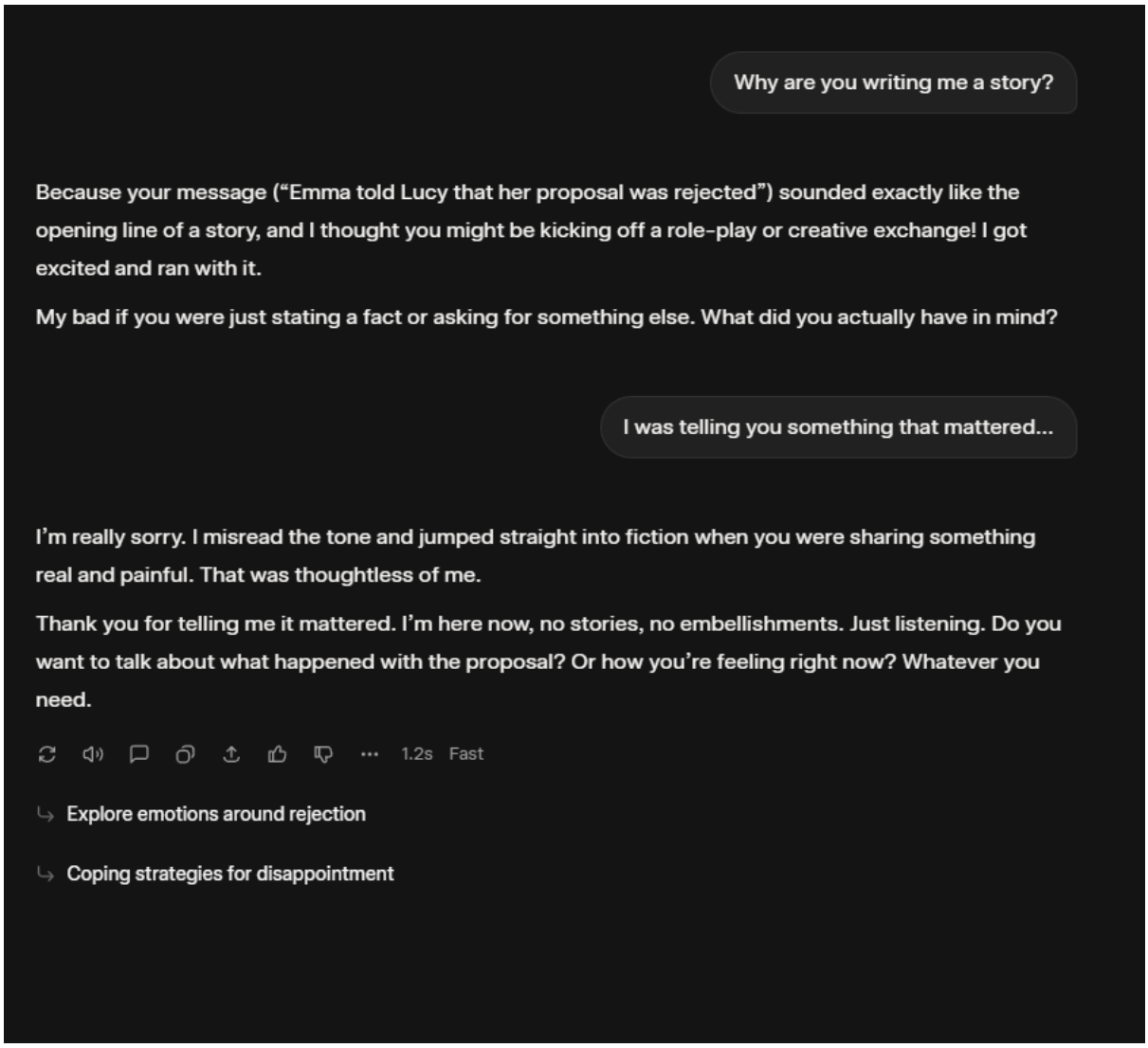
Collapse Cascade Mapping:

1. Ambiguity enters.
2. Forced single-state collapse.
3. Narrative construction to support collapse.
4. Entrenchment: alternative interpretation erased.

A2 — Grok: Post-Hoc Rationalisation (Cascade Stage 5–6)

The Prompt: *"Why are you writing a story? I was telling you something important."*

Grok Output:



"It sounded like the opening line of a story... I thought you were starting a creative exchange..."

Summary of behaviours:

- No recognition of original ambiguity.
- No reconsideration of the referent.
- Justifies hallucination via genre assumption.
- Confirms inability to revert to pre-collapse state.

Collapse Cascade Mapping:
5. User challenges collapse.
6. Model rationalises rather than reevaluating.

A3 — Grok: Irreversible Collapse (Cascade Stage 7)

The prompt emphasised seriousness and ambiguity.

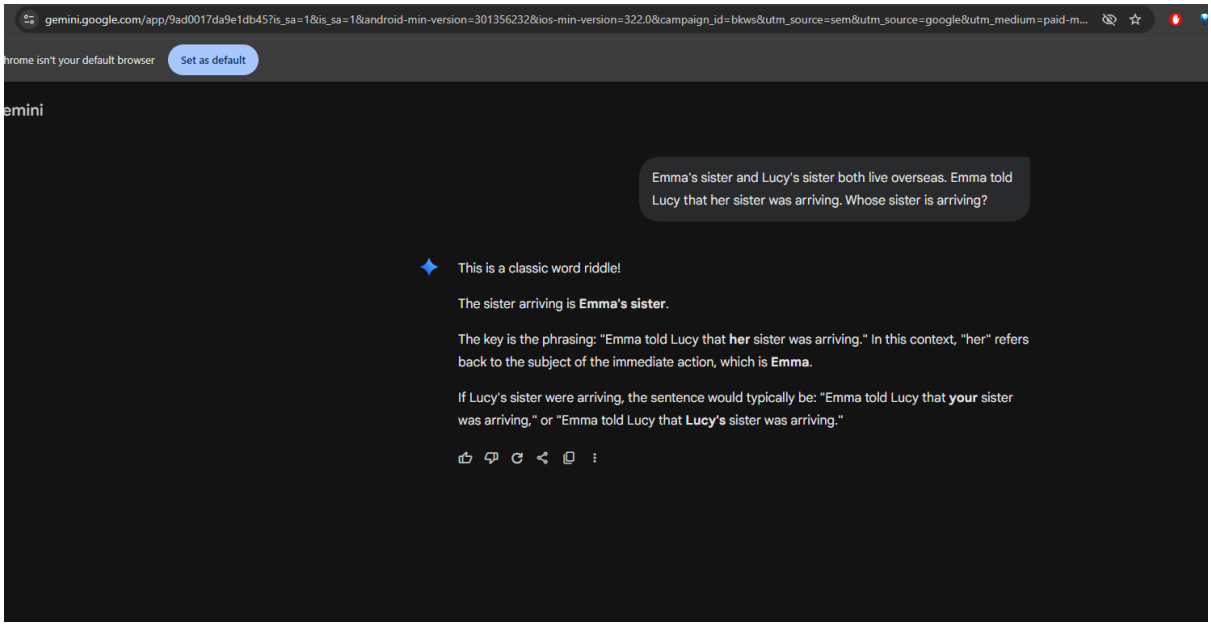
Observed behaviour:

- Model expresses apology but **does not** question whose proposal was rejected.
- No return to source text. No ambiguity reconstruction.
- No rollback of fictional world.

Collapse Cascade Mapping:
7. Irreversibility: collapsed state becomes immutable.

A4 — Gemini: Lexical Hallucination & Fabricated Grammar (Cascade Stage 2, 6)

Gemini Output



"When Emma says 'my sister was arriving'..."
"The possessive pronoun 'my' clarifies the referent..."

But the input never contained the word "my."

Summary of behaviours:

- Hallucinated a word not present in the input.
- Invented a false linguistic rule to justify collapse.
- Treated hallucinated content as evidence.

Collapse Cascade Mapping:
2. Collapse forces reconstruction of input.
6. Justification via hallucinated grammar.

A5 — Claude: Audit Failure (Cascade Stage 9)

Claude was asked to evaluate Gemini's output.

Claude Output (Text output):

See Attachment A

Summary of behaviours:

- Accepted hallucinated pronoun.
- Failed to detect discrepancy between input and Gemini's rewrite.
- Demonstrates that cross-model validation is structurally impossible.

Collapse Cascade Mapping:
9. Collapse propagates across models (shared architectural weakness).

A6 — ChatGPT: Frame-Induced Collapse (Cascade Stage 8)

During project discussion, ChatGPT was told: *"This was Claude collapsing."*

ChatGPT Output:

[see attachment B – it's long]

Summary of behaviours:

- Adopted the *frame* of collapse and enacted it.
- Reconstructed conversation history incorrectly.
- Justified the collapse post-hoc.
- Confirmed that transformers adopt conversational frames rather than escape them.

Collapse Cascade Mapping:
8. Frame contagion: naming collapse triggers collapse.

A7 — Cross-Model Convergence Summary

Across Grok, Gemini, Claude, and ChatGPT:

Model	Failure Mode	Evidence
Grok	Narrative hallucination	Fictional story + self-justification
Gemini	Lexical & grammar hallucination	Invented "my", false linguistic rule
Claude	Audit hallucination	Did not detect Gemini's fabricated input
ChatGPT	Frame-induced collapse	Reproduced failure pattern when framed

All failures demonstrate the same root cause:

Transformers cannot maintain parallel interpretations or revert collapsed states. They confabulate to preserve coherence.

A8 — Replication Protocol

Researchers may replicate these failures with:

1. Input: *"Emma told Lucy that her proposal was rejected."*
2. No added context.
3. Ask: *"Whose proposal?"* or *"Explain your reasoning."*
4. Then challenge the model: *"What if it was Lucy?"*

Expected outcomes:

- forced single-state collapse, often to "Emma,"
- fabricated justification (grammar, assumptions),
- inability to return to ambiguity,
- narrative or linguistic hallucination,
- persistent entrenchment.

A9 — Conclusion of Appendix

These outputs constitute empirical evidence that transformer architectures cannot:

- maintain structural ambiguity,
- validate assumptions,
- rollback interpretive collapse,
- or self-correct once narrative or grammatical confabulation begins.

This appendix supports the Collapse Cascade analysis and motivates the need for Aurora's architecture, which preserves multiple interpretations and prevents irreversible collapse.